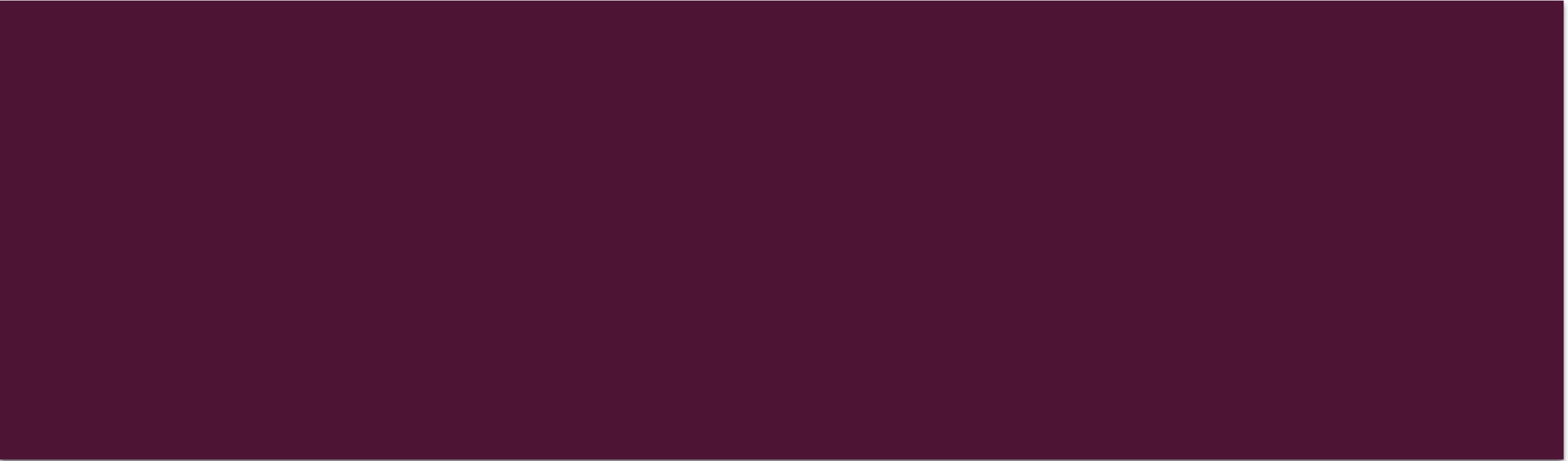




CHAPTER 7 CONTINUES



REVIEW QUESTION I

- What is the difference between an accumulator and a counter?

REVIEW QUESTION | ANSWER

- A counter counts. Like 1, 2, 3, 4.
- An accumulator accumulates like $10+15+30$.

REVIEW QUESTION 2

- If you have a **priming read** before a loop condition, but no **update read**, what will be the result?

REVIEW QUESTION 2 ANSWER

- The variable controlling the loop will never be updated.
- This will cause an endless loop.

REVIEW QUESTION 3

- What is the purpose of the **sentinel value** and where does it come from?

REVIEW QUESTION 3 ANSWER

- The sentinel value is a value that ends a loop. It can be any value that is logical for ending the loop.

REVIEW QUESTION 4

- The instructions between the brackets after a while statement are known as the _____?

REVIEW QUESTION 4 ANSWER

- They are the loop body.

REVIEW QUESTION 5

- Why do you use a ***break*** statement in a switch/case statement?

REVIEW QUESTION 5 ANSWER

- The break statement ends the processing for that value. It can shorten the processing time for the code because every case will not need to be evaluated.

REVIEW QUESTION 6

- What is the following structure called?
- ```
if (age > 0)
```

  - ```
{
```
 - ```
cout << "Your age is:" << age << endl;
```
  - ```
if ( age == 99)
```

 - ```
Cout << "You are almost 100!" << endl;
```
    - ```
}//endif
```

REVIEW QUESTION 6 ANSWER

- A single-decision structure.

NEW STATEMENT

- **for**
 - Mostly for counter-controlled pretest loops
 - `for (int x = 1; x<4; x+=1)`
 - `cout << x << endl;`
 - `//end for`

PRACTICE I

- A program declares an int variable named evenNum and initializes it to 2. Write a C++ while loop that uses the evenNum variable to display the even integers between 1 and 9.

ANSWER 1

- `int evenNum = 2;`
- `while (evenNum < 9)`
- `{`
 - `cout << "The number is:" << evenNum;`
 - `evenNum = evenNum + 2;`
 - `//endwhile`

QUESTION ABOUT *FOR*?

- Which of the following **for** clauses processes the loop instructions as long as the `x` variable's value is less than or equal to the number 100?
- `for (int x = 10; x <= 100; x = x + 10)`
- `for (int x = 10; x < 100, x = x + 10)`
- `for (int x == 10; x <= 100; x = x + 10)`
- `for (int x = x + 10; x <= 100; x = 10)`

ANSWER ABOUT FOR

- `for (int x = 10; x <= 100; x = x + 10)`

QUESTION ABOUT LOOPS

- The computer will stop processing the loop associated with the **for** clause from the previous question when the `x` variable contains the number -----?
- `for (int x = 10; x <= 100; x = x + 10)`
 - a. 100
 - b. 111
 - c. 101
 - d. 110

QUESTION ABOUT LOOPS ANSWER

■ The computer will stop processing the loop associated with the **for** clause from the previous question when the x variable contains the number -----?

a. 100

b. 111

c. 101

d. 110

FINAL QUESTION

- Write a **for** clause that processes the loop instructions as long as the value stored in the x variable is greater than the number 0. The x variable should be an int variable. Initialize the variable to the number 25 and update it by -5 with each repetition of the loop.

ANSWER: HOW CLOSE WERE YOU?

- `for (int x = 25; x > 0; x = x-5)`

QUESTION OF THE DAY

- Name the three programming structures you have learned about in this book.

ANSWER OF THE DAY

- Sequence
- Selection
- Repetition